

Zara Williams

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Professional Summary

- Embedded Software Engineer with 4 years of experience developing firmware for connected devices and sensor systems.
- Skilled in C, C++, RTOS concepts, microcontrollers, communication protocols, and hardware debugging tools.
- Experienced in writing reliable low-level software, supporting board bring-up, and validating device behavior.

Skills

Languages:

C, C++, Python

Embedded:

Firmware Development, Microcontrollers, RTOS, Device Drivers

Protocols:

UART, SPI, I2C, BLE, MQTT

Hardware:

Oscilloscope, Logic Analyzer, PCB Bring-up

Tools:

STM32CubeIDE, Keil, Git, CMake

Work Experience

Embedded Software Engineer at SensorForge Technologies

Liverpool, UK | Sep 2022 - Present

- Developed C firmware for environmental sensor devices using STM32 microcontrollers.
- Implemented BLE configuration workflows and MQTT telemetry publishing.
- Improved device reliability by adding watchdog handling, fault states, and diagnostic logs.

Firmware Developer at NorthCircuit Labs

Manchester, UK | Jul 2020 - Aug 2022

- Built low-level drivers for SPI sensors, I2C peripherals, and UART-based modules.
- Supported hardware bring-up, firmware flashing, and field issue analysis.
- Created Python scripts for automated device testing and serial log collection.

Projects

AirSense IoT Node

Tech Stack: C, STM32, BLE, MQTT

Built firmware for an environmental monitoring node with sensor sampling, BLE setup, and cloud telemetry.

Device Diagnostics Harness

Tech Stack: Python, UART, CMake, CSV

Created an automated diagnostics workflow for validating firmware logs, sensor readings, and fault states.

Education

B.Eng. in Electronics and Embedded Systems - University of Liverpool

Liverpool, UK | 2016 - 2020

Certifications

- Embedded C Programming Certification
- Bluetooth Low Energy Development Certification